Agency Exercises (V2): Solutions

Agent Foundations module
Iliad Intensive

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Notation follows the exercise sheet: L is a fixed consistent formal system, $L \vdash \varphi$ means φ is provable in L, K is a fixed contradiction (so consistency is $L \nvdash K$), and $\Box P$ is the statement, internal to L, that "P is provable", equivalently that $\mathsf{ProofSeeker}(P)$ halts. The bridge: if a program halts then L proves it halts; if $L \vdash P$ then $\mathsf{ProofSeeker}(P)$ halts.

1 Exercise 1: Gödel's second incompleteness theorem

Recall $Z(A)$ searches for an L-proof of $\neg\mathsf{Halts}(A(A))$ and halts if it finds one, and

$$G \;:=\; \neg\mathsf{Halts}(Z(Z)).$$

By construction $Z(Z)$ searches for an L-proof of G, so

$$Z(Z) \text{ halts} \iff L \vdash G. \tag{$\star$}$$

Part 1(a). G is true (assuming L consistent).

Solution. Two cases.

- *$Z(Z)$ halts.* By ($\star$), $L \vdash G$, i.e. $L \vdash \neg\mathsf{Halts}(Z(Z))$. But $Z(Z)$ actually halts, so by the bridge $L \vdash \mathsf{Halts}(Z(Z))$. Then L proves both $\mathsf{Halts}(Z(Z))$ and its negation, contradicting consistency.
- *$Z(Z)$ runs forever.* Then $\neg\mathsf{Halts}(Z(Z))$, i.e. G, is true.

Consistency rules out the first case, so $Z(Z)$ runs forever and G is true. $\Box$

Part 1(b). If $L \vdash \neg\Box K$ (i.e. L proves its own consistency) then L is inconsistent.

Solution. The argument of 1(a) used only the consistency of L, and each step is a finite manipulation of programs and proofs that L can formalize. Reading it inside L: "$Z(Z)$ halts" yields $\Box G$ (by construction) and $\Box\mathsf{Halts}(Z(Z))$ (the bridge), hence $\Box K$; contrapositively $L \vdash \neg\Box K \to G$. If $L \vdash \neg\Box K$, modus ponens gives

$$L \vdash G, \qquad \text{i.e.} \qquad L \vdash \neg\mathsf{Halts}(Z(Z)).$$

But $L \vdash G$ means $Z(Z)$'s search finds a proof of G, so $Z(Z)$ halts; by the bridge $L \vdash \mathsf{Halts}(Z(Z))$. Now L proves both $\mathsf{Halts}(Z(Z))$ and $\neg\mathsf{Halts}(Z(Z))$, so L is inconsistent. $\Box$

Part 1(c). If $L \vdash \Box P \to P$ for every P, then L is inconsistent.

Solution. (i) Fix P with $L \vdash \neg P$. The contrapositive of $\Box P \to P$ is $\neg P \to \neg\Box P$, so from $L \vdash \Box P \to P$ we get $L \vdash \neg P \to \neg\Box P$. With $L \vdash \neg P$, modus ponens gives $L \vdash \neg\Box P$.

(ii) Take $P = K$. Since $\neg K$ is a tautology, $L \vdash \neg K$, so by (i) $L \vdash \neg\Box K$. Thus L proves its own consistency, and Part 1(b) makes L inconsistent. $\Box$

2 Exercise 2: Löb's theorem

We use Necessitation (N) $L \vdash \varphi \Rightarrow L \vdash \Box\varphi$, Distribution (K) $L \vdash \Box(\varphi \to \psi) \to (\Box\varphi \to \Box\psi)$, and the Löb condition (4) $L \vdash \Box\varphi \to \Box\Box\varphi$.

Part 2(a). Necessitation. If $L \vdash \varphi$ there is a concrete proof of φ; searching all strings, $\mathsf{ProofSeeker}(\varphi)$ meets it and halts. "$\mathsf{ProofSeeker}(\varphi)$ halts" is exactly $\Box\varphi$, and a halting run is finite, so by the bridge $L \vdash \Box\varphi$. $\Box$

Part 2(b). Distribution. A proof of $\varphi \to \psi$ turns a proof of φ into one of ψ (concatenate the two proofs and apply modus ponens). So if $\mathsf{ProofSeeker}(\varphi \to \psi)$ and $\mathsf{ProofSeeker}(\varphi)$ both halt, their outputs combine into a proof of ψ, whence $\mathsf{ProofSeeker}(\psi)$ halts. This combining is a finite procedure L can carry out, so $L \vdash \Box(\varphi \to \psi) \to (\Box\varphi \to \Box\psi)$. $\Box$

The Löb sentence λ satisfies $L \vdash \lambda \leftrightarrow (\Box\lambda \to C)$.

Part 2(c). $L \vdash \Box\lambda \to \Box C$.

Solution. From $L \vdash \lambda \to (\Box\lambda \to C)$:

$$\begin{aligned}
&\text{(N):} \quad L \vdash \Box\big(\lambda \to (\Box\lambda \to C)\big),\\
&\text{(K):} \quad L \vdash \Box\lambda \to \Box(\Box\lambda \to C),\\
&\text{(K):} \quad L \vdash \Box(\Box\lambda \to C) \to (\Box\Box\lambda \to \Box C),
\end{aligned}$$

and chaining the last two, $L \vdash \Box\lambda \to (\Box\Box\lambda \to \Box C)$. By (4), $L \vdash \Box\lambda \to \Box\Box\lambda$. Propositionally, from $\Box\lambda$ we obtain $\Box\Box\lambda$ and then $\Box C$, so $L \vdash \Box\lambda \to \Box C$. $\Box$

Part 2(d). Assuming $L \vdash \Box C \to C$, derive $L \vdash C$.

Solution. Chaining $L \vdash \Box\lambda \to \Box C$ (2(c)) with $L \vdash \Box C \to C$ gives $L \vdash \Box\lambda \to C$. The Löb equivalence gives the converse direction $L \vdash (\Box\lambda \to C) \to \lambda$, so $L \vdash \lambda$. By (N), $L \vdash \Box\lambda$; with $L \vdash \Box\lambda \to C$, modus ponens yields $L \vdash C$. $\Box$

Part 2(e). FairBot. With $A :=$ "$\mathsf{FairBot}_1(\mathsf{FairBot}_2) = C$" and $B :=$ "$\mathsf{FairBot}_2(\mathsf{FairBot}_1) = C$", show $L \vdash A \wedge B$.

Solution. From the source code, $\mathsf{FairBot}_1$ returns C iff it finds an L-proof that its opponent $\mathsf{FairBot}_2$ returns C against it, i.e. a proof of B; reading this off the code,

$$L \vdash \Box B \to A, \qquad L \vdash \Box A \to B.$$

For any φ, $\Box(A \wedge B) \to \Box\varphi$ whenever $A \wedge B \to \varphi$ is a tautology: indeed (N) gives $\Box(A \wedge B \to \varphi)$ and (K) then gives $\Box(A \wedge B) \to \Box\varphi$. Applying this to $\varphi = A$ and $\varphi = B$,

$$L \vdash \Box(A \wedge B) \to \Box A, \qquad L \vdash \Box(A \wedge B) \to \Box B.$$

Hence, assuming $\Box(A \wedge B)$: we get $\Box A$ and $\Box B$, then B (from $\Box A \to B$) and A (from $\Box B \to A$), so $A \wedge B$. Thus

$$L \vdash \Box(A \wedge B) \to (A \wedge B),$$

and Löb's theorem with $C := A \wedge B$ gives $L \vdash A \wedge B$. $\Box$

3 Exercise 3: The Complete Class Theorem

Higher reward is better; δ' *dominates* δ if $r(\delta') \geq r(\delta)$ coordinatewise with strict inequality in some coordinate. $r(\delta)$ is linear in the mixing weights, $R = \operatorname{conv}\{r(a_1), \ldots, r(a_k)\}$, and $\mathrm{EU}(\delta, \pi) = \pi \cdot r(\delta)$.

Part 3(a)(i). An admissible $\delta = \sum_j \lambda_j a_j$ puts $\lambda_j = 0$ on every non-Pareto-optimal (dominated) a_j.

Solution. Suppose $\lambda_j > 0$ for a dominated a_j, and let δ' dominate it: $r(\delta') \geq r(a_j)$ with strict inequality in some coordinate i_0. Replace the weight on a_j by δ', i.e. play δ' with the probability λ_j formerly on a_j. This is a valid rule $\hat{\delta}$ with

$$r(\hat{\delta}) = r(\delta) + \lambda_j\big(r(\delta') - r(a_j)\big) \geq r(\delta),$$

and strict in coordinate i_0 (since $\lambda_j > 0$). So $\hat{\delta}$ dominates δ, contradicting admissibility. Hence $\lambda_j = 0$, i.e. $F \subseteq \operatorname{conv}\{\text{Pareto-optimal pure actions}\}$. $\Box$

Part 3(a)(ii). Every Bayes-optimal rule under a prior π with $\pi_i > 0$ for all i is admissible.

Solution. If δ' dominated such a δ, then with $r(\delta')_{i_0} > r(\delta)_{i_0}$,

$$\mathrm{EU}(\delta', \pi) - \mathrm{EU}(\delta, \pi) = \sum_i \pi_i\big(r(\delta')_i - r(\delta)_i\big) \geq \pi_{i_0}\big(r(\delta')_{i_0} - r(\delta)_{i_0}\big) > 0,$$

since every term is ≥ 0 and the i_0 term is > 0. This contradicts Bayes-optimality, so δ is admissible. $\Box$

Part 3(b). For a face $\mathcal{F} = \{\sum_{l=1}^{p} \mu_l\, r(a_{i_l}) : \mu_l \geq 0,\ \sum_l \mu_l = 1\}$ with $v_l := r(a_{i_l}) - r(a_{i_1})$ and $H = \operatorname{span}\{v_2, \ldots, v_p\}$: H is exactly the set of directions tangent to $\mathcal{F}$.

Solution. Any point of $\mathcal{F}$ equals $r(a_{i_1}) + \sum_{l \geq 2} \mu_l v_l$, so $\mathcal{F} \subseteq r(a_{i_1}) + H$. Take $x, x + h \in \mathcal{F}$, say $x = \sum_l \mu_l r(a_{i_l})$ and $x + h = \sum_l \mu'_l r(a_{i_l})$ with $\sum_l \mu_l = \sum_l \mu'_l = 1$. Then

$$h = \sum_l (\mu'_l - \mu_l)\, r(a_{i_l}) = \sum_{l \geq 2} (\mu'_l - \mu_l)\, v_l \in H,$$

using $\sum_l (\mu'_l - \mu_l) = 0$ to eliminate the $r(a_{i_1})$ term. Hence moving within $\mathcal{F}$ requires $h \in H$. Conversely, from any relative-interior point ($\mu_l > 0$) every $h \in H$ is realized: $x + \varepsilon h \in \mathcal{F}$ for small $\varepsilon > 0$. So the tangent directions of $\mathcal{F}$ are precisely H. $\Box$

Part 3(c). Let $\pi \perp H$ be normalized so $\pi_i > 0$ and $\sum_i \pi_i = 1$, and assume no point of R scores strictly higher under π than the points of $\mathcal{F}$.

Solution. (i) For $x, x' \in \mathcal{F}$ we have $x - x' \in H$ by 3(b), so $\pi \cdot (x - x') = 0$. Thus $\pi \cdot r(\delta)$ equals a common value c for all δ with $r(\delta) \in \mathcal{F}$.

(ii) Let $r(\delta) \in \mathcal{F}$, so $\mathrm{EU}(\delta, \pi) = c$. For any δ', $r(\delta') \in R$, and the supporting assumption gives $\pi \cdot r(\delta') \leq c$. Hence $\mathrm{EU}(\delta', \pi) \leq c = \mathrm{EU}(\delta, \pi)$, so δ is Bayes-optimal under π.

Conclusion. If δ is admissible then $r(\delta) \in F$ lies on some face $\mathcal{F}$; the prior π built from $\mathcal{F}$ has all $\pi_i > 0$ and, by (ii), makes δ Bayes-optimal. Conversely, by 3(a)(ii) every Bayes-optimal rule under a strictly positive prior is admissible. The two classes coincide. □

4 Exercise 4: The Do-Divergence Theorem

Write $\Pr[X, A, O] = \Pr[X \mid A, O] \Pr[A \mid O] \Pr[O]$ and the blind baseline $\Pr[X, A, O \mid \mathrm{do}(A)] = \Pr[X \mid A, O] \Pr[A] \Pr[O]$.

Claim. $D_{\mathrm{KL}}\big(\Pr[X] \,\big\|\, \Pr[X \mid \mathrm{do}(A)]\big) \leq \mathrm{MI}(A; O)$.

Solution. Compute the divergence between the full joints. The factors $\Pr[X \mid A, O]$ and $\Pr[O]$ cancel in the log-ratio:

$$D_{\mathrm{KL}}\big(\Pr[X, A, O] \,\big\|\, \Pr[X, A, O \mid \mathrm{do}(A)]\big) = \sum_{x,a,o} \Pr[x, a, o] \log \frac{\Pr[a \mid o]}{\Pr[a]}.$$

The summand depends only on a, o; marginalizing x and using $\Pr[a \mid o] / \Pr[a] = \Pr[a, o] / (\Pr[a] \Pr[o])$,

$$= \sum_{a,o} \Pr[a, o] \log \frac{\Pr[a, o]}{\Pr[a] \Pr[o]} = \mathrm{MI}(A; O).$$

Marginalizing the two joints down to X sends them to $\Pr[X]$ and $\Pr[X \mid \mathrm{do}(A)] = \sum_{a,o} \Pr[X \mid a, o] \Pr[a] \Pr[o]$ respectively, so monotonicity of KL under marginalization gives

$$D_{\mathrm{KL}}\big(\Pr[X] \,\big\|\, \Pr[X \mid \mathrm{do}(A)]\big) \leq D_{\mathrm{KL}}\big(\Pr[X, A, O] \,\big\|\, \Pr[X, A, O \mid \mathrm{do}(A)]\big) = \mathrm{MI}(A; O).$$

5 Exercise 5: Channel Additivity

The channel factorizes as $P[Y \mid X] = P[Y_1 \mid X_1] P[Y_2 \mid X_2]$, and $P[Y, X] = P[Y \mid X] P[X]$.

Part 5(a). $\mathrm{MI}(X; Y) = \mathrm{MI}(X_1; Y_1) + \mathrm{MI}(X_2; Y_2) - \mathrm{MI}(Y_1; Y_2)$.

Solution. Using $P[x, y] = P[y \mid x] P[x]$ and the factorization,

$$\mathrm{MI}(X; Y) = \sum_{x,y} P[x, y] \log \frac{P[y \mid x]}{P[y]} = \sum_{x,y} P[x, y] \log \frac{P[y_1 \mid x_1] P[y_2 \mid x_2]}{P[y_1, y_2]}.$$

The three terms on the right are, respectively,

$$\mathrm{MI}(X_1; Y_1) = \sum P[x, y] \log \tfrac{P[y_1 | x_1]}{P[y_1]}, \quad \mathrm{MI}(X_2; Y_2) = \sum P[x, y] \log \tfrac{P[y_2 | x_2]}{P[y_2]},$$

$$\mathrm{MI}(Y_1;Y_2) = \sum P[x,y] \log \tfrac{P[y_1,y_2]}{P[y_1]P[y_2]},$$

each written over the full joint (the first depends only on x_1, y_1, etc.). Then

$$\mathrm{MI}(X_1;Y_1) + \mathrm{MI}(X_2;Y_2) - \mathrm{MI}(Y_1;Y_2) = \sum_{x,y} P[x,y] \log \frac{P[y_1 \mid x_1]P[y_2 \mid x_2]}{P[y_1,y_2]} = \mathrm{MI}(X;Y). \quad \square$$

Part 5(b). For $Q[X_1,X_2] := P[X_1]P[X_2]$, $\mathrm{MI}_Q(X;Y) \geq \mathrm{MI}_P(X;Y)$.

Solution. Q keeps the marginals $Q[X_i] = P[X_i]$ and the channel, so $\mathrm{MI}_Q(X_i;Y_i) = \mathrm{MI}_P(X_i;Y_i)$ for $i = 1,2$ ($\mathrm{MI}(X_i;Y_i)$ depends only on $P[X_i]$ and $P[Y_i \mid X_i]$). Under Q the inputs are independent, so the outputs are too:

$$Q[y_1,y_2] = \sum_{x_1,x_2} Q[x_1]Q[x_2]P[y_1 \mid x_1]P[y_2 \mid x_2] = Q[y_1]\,Q[y_2],$$

hence $\mathrm{MI}_Q(Y_1;Y_2) = 0$. Applying 5(a) under each distribution,

$$\mathrm{MI}_Q(X;Y) = \mathrm{MI}_P(X_1;Y_1) + \mathrm{MI}_P(X_2;Y_2) = \mathrm{MI}_P(X;Y) + \mathrm{MI}_P(Y_1;Y_2) \geq \mathrm{MI}_P(X;Y),$$

the last step using $\mathrm{MI}_P(Y_1;Y_2) \geq 0$. $\square$

Part 5(c). A throughput-maximizing input with $X_1 \perp X_2$ always exists.

Solution. The input simplex is compact and $\mathrm{MI}(X;Y)$ continuous, so a maximizer P^* exists. Let $Q^* := P^*[X_1]P^*[X_2]$. By 5(b), $\mathrm{MI}_{Q^*}(X;Y) \geq \mathrm{MI}_{P^*}(X;Y) = \max$, so Q^* is also a maximizer, and under Q^* the inputs are independent. $\square$